

Day 8: Chrome/Edge MV3 Extension Architecture & In-Gmail Shield

Key Milestone: Browser Extension Manifest V3 & In-Page Injection

Sprint Focus: Execution of core deliverables for Day 8 across AI/NLP, Header Forensics, IOC Extraction, Graph Correlation, Case Management, and UI Integration.

Daily Deliverables: MV3 Extension Package, Gmail DOM Ingestion Script, Extension Popup Controller.

Individual Team Member Task Breakdown & Contributions

Team Member & Lead Role	Domain & Ownership Area	Tasks Executed & Deliverables on Day 8
Muniswami.Y AI / NLP Lead	Explainable AI, Intent & Phishing Detection, Social Engineering signals	Optimized lightweight heuristic model to run client-side in extension background worker for zero-latency offline threat scoring.
Jeevan reddy.G Header & Forensics Lead	Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis	Engineered DOM parser in `content.js` to extract raw email headers, sender display names, and recipient addresses from Gmail UI.
Lahari.K IOC & Infrastructure Lead	URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection	Built real-time URL link scanner inside `content.js`: inspecting all hyperlink anchor tags in email bodies for deceptive display-vs-href mismatches.
Hemanth.P Graph & Campaign Intelligence Lead	Attack Relationship Graph, Similarity detection, Campaign grouping & correlation	Designed lightweight payload compressor for transfer between Extension Service Worker and Frontend Cockpit via URL hash state.
Abbas.N Risk, Case & Response Lead	Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions	Created extension message router handling `ANALYZE_CURRENT_TAB`, `REPORT_CYBERCRIME`, and `GENERATE_CANARY` actions.
Moulaanbee.N Frontend + Assistant + Simulator Lead	Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX	Designed and coded the Extension Popup interface (`popup.html` / `popup.js`) and in-Gmail floating forensic score shield overlay.

Integration, Verification & Quality Assessment

Cross-Module Integration:	Successfully loaded unpacked extension in Chrome & Edge; verified automatic threat scoring badge injection inside Gmail.
Quality Assurance Status:	All code unit tests and interface contracts for Day 8 verified clean. Git commit pushed with zero blocking errors.